

of flyby missions, starting all the way from the Pioneer series. Pioneer 10 flew past Jupiter in 1973, and Pioneer 11 flew past it only a year after in 1974. The former collected information on its atmosphere, magnetic fields and radiation while the latter shot the first

indirect way of measuring water) and Jupiter's gravitational and magnetic fields among other things. Oh also, just to get your trivia game up – NASA partnered up with LEGO to send figurines to Jupiter! LEGO figurines of the Roman God Jupiter, his wife Juno and Galileo

future expansion. Europa Clipper is a proposed mission that shares many similarities with JUICE. It's launching around 2022 as well and its primary objective is to obtain an indepth study of Europa via a lander. It'll also look for key compositions and geology and look for organic compounds. It is meant as a complement to JUICE and both missions hold great promise for human habitation studies. Right now the biggest problems are radiation and lack of knowledge of the system.

Y'ALL GOT ANY MORE OF THOSE MOONS?

Jupiter recently just had two more moons added to its already impressive arsenal of 67, bumping the total upto 69 (Takes on a whole new light when you consider most of Jupiter's moons were named after his wives and conquests). What's more, the scientists responsible for this discovery also found several 'lost' moons! Since Jupiter is so large, (318 Earth masses, remember?) it naturally has an gravitational order of magnitude several orders higher than that of Earth. So this allows it to have a large gravitational field surrounding it which in turn leads to more satellites orbiting it and at very huge distances as well. For example, the two most newest moons, S/2016 J1 and S/2017 J1 are 21 and 24 million kilometers away from Jupiter respectively! For comparison, our moon is only 384,400 km away! And as for lost moons – some of Jupiter's moons have orbits so long that after they've been spotted and studied once, people weren't able to find them again! The scientists over at the Department of Terrestrial Magnetism at Carnegie Science were collecting data on the side on Jupiter while they were busy searching for Planet X. When they compared a whole year's worth of data from 2016-2017 to data previously collected in 2003, they found two completely new moons as well five moons that were previously considered lost! This brings down the list of lost moons down to nine from fourteen, and provides even more insight into the whole Jovian system. Maybe there are more moons like the aforementioned Galilean moons which can support human habitation? Read more at <http://digit.in/Jupiter69> 



This honestly looks more like a ceiling fan



The Galilean Moons are a pretty bunch

actual close ups of the planet we would ever see. The data gathered by these two spacecrafts was vital in figuring out how to make better ones to further study Jupiter, and there have been 5 other such 'flyby' missions after them upto 'New Horizon' – a Pluto headed probe that was launched in 2007. The first orbiting spacecraft for Jupiter was the Galileo (He does deserve the honour for discovering the planet among other things) in 1995, over twenty years since the first flyby mission. Galileo ended up orbiting Jupiter for over seven years, gathering immense amounts of data all the while and it was decommissioned by a 'controlled collision with the Jovian surface'. The next orbiter mission was Juno which was launched on August 5, 2011. It finally descended into a polar orbit around Jupiter on July 5, 2016 and is now busy measuring the ratio of hydrogen to oxygen (which is just an

Galilei are aboard the spacecraft Juno. Obviously the regular plastic LEGOs we are used to won't make the trip very well, so these are made of a special aircraft grade aluminum.

There are currently two missions planned for Jupiter in the future : European Space Agency's Jupiter Icy Moon Explorer or JUICE (We can't even...) and NASA's Europa Clipper. JUICE is supposed to launch around 2022 and hopefully make it close to Jupiter at around 2030. Its primary objective is to study three out of four Galilean moons (Moons that were discovered by Galileo along with Jupiter) – Ganymede, Callisto and Europa. All of these moons are suspected to have liquid water on their surface, and all of them, especially Europa, could be a potentially habitable place for humankind. JUICE is going to be the first 'subsurface sounding' we get of Europa, so this is good news for